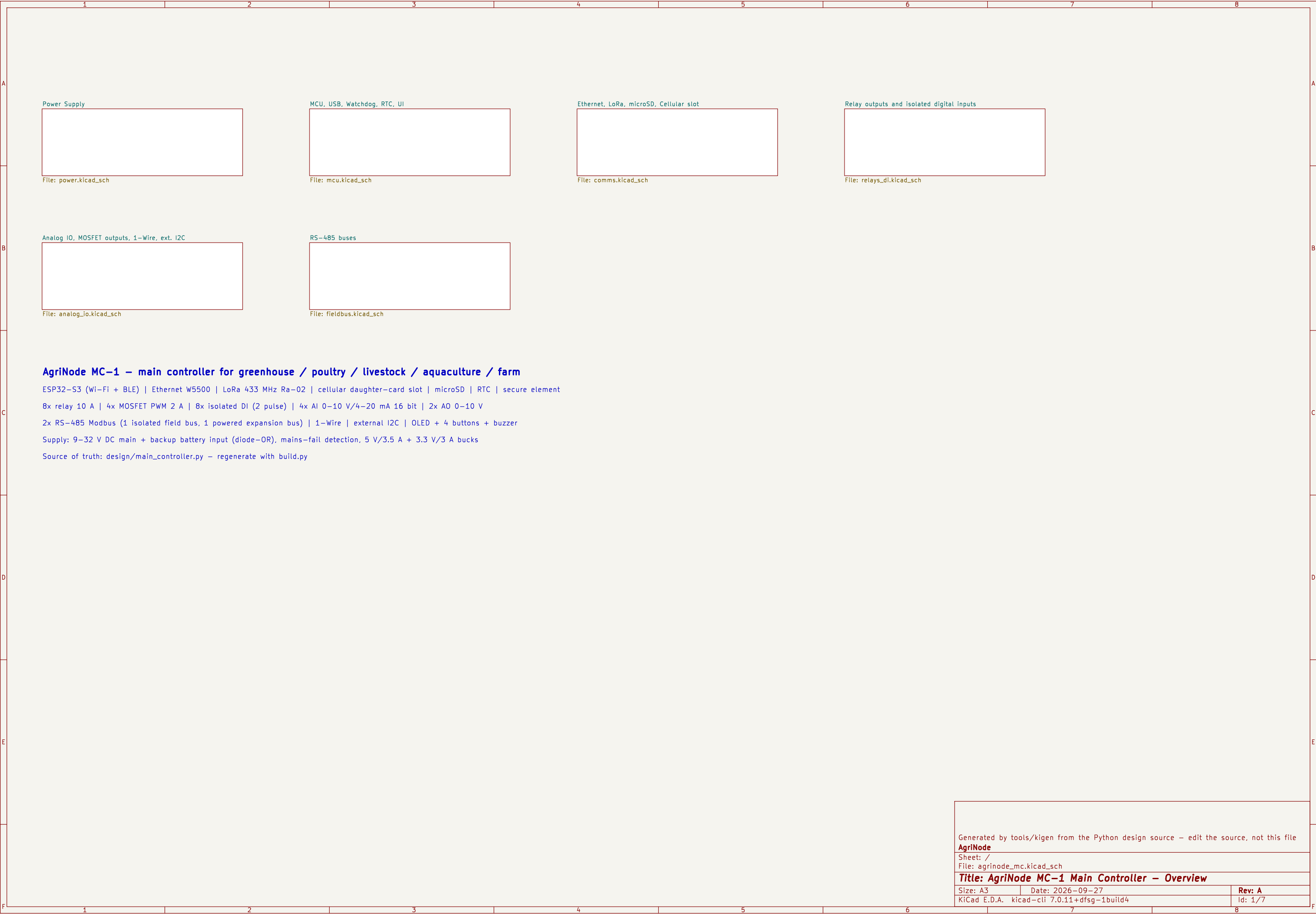
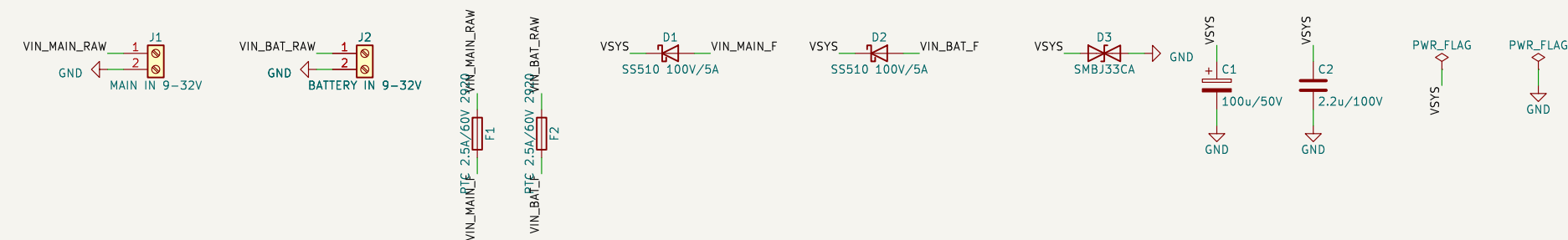
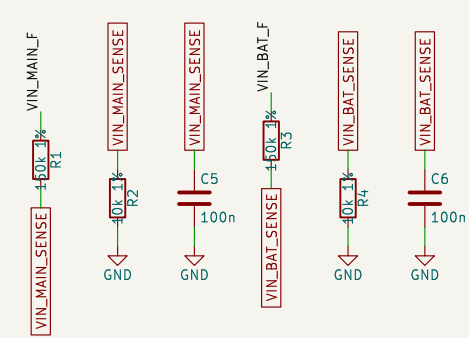




9–32 V DC; each input fused; diode-OR also gives reverse-polarity protection

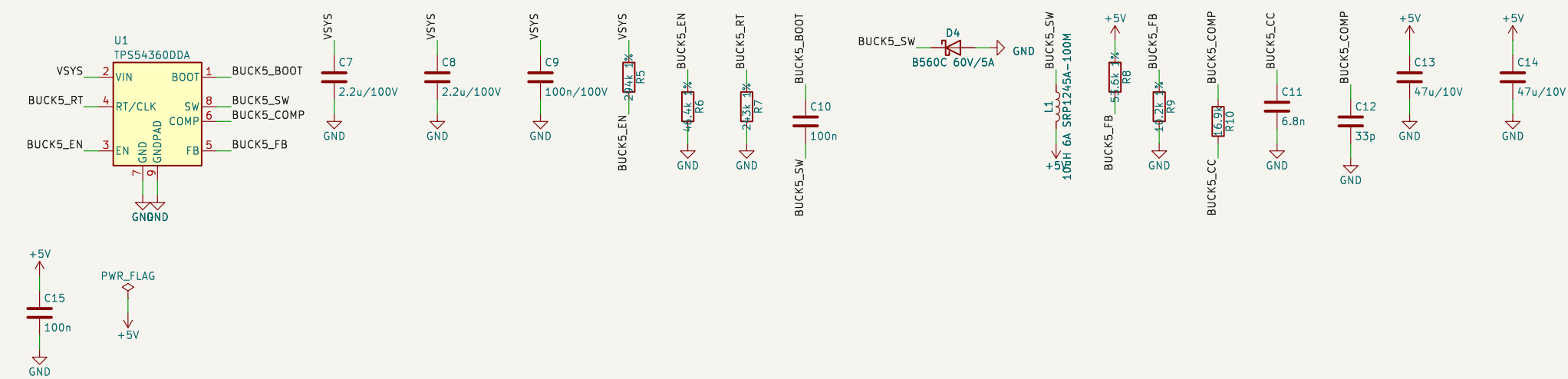
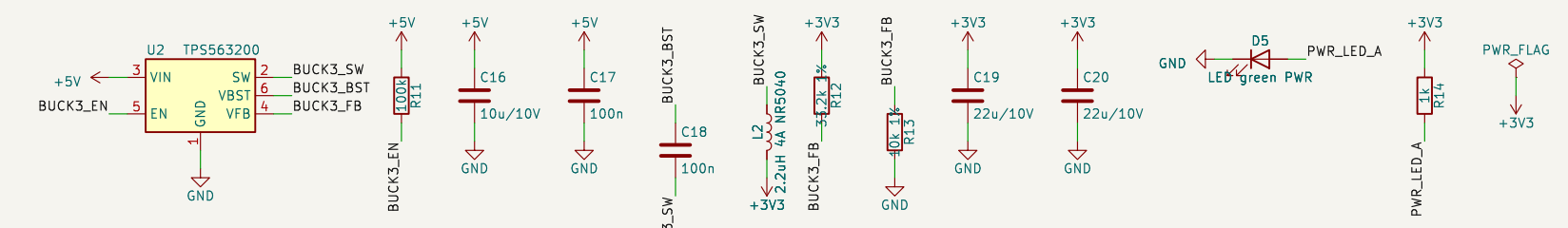


150k/10k: 32 V $\rightarrow$ 2.0 V

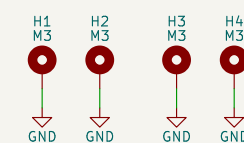
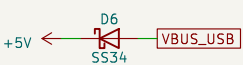


fsw 400 kHz, UVLO on 8.5 V / off 7.5 V

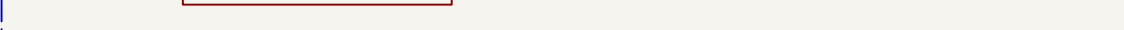
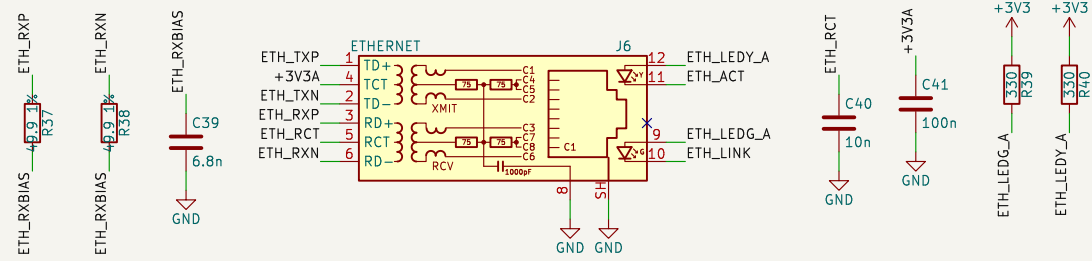
fsw 400 kHz, UVLO on 8.5 V / off 7.5 V


$$V_{out} = 0.768 \times (1 + 33.2k/10k) = 3.32 \text{ V}$$
$$V_{out} = 0.768 \times (1 + 33.2k/10k) = 3.32 \text{ V}$$


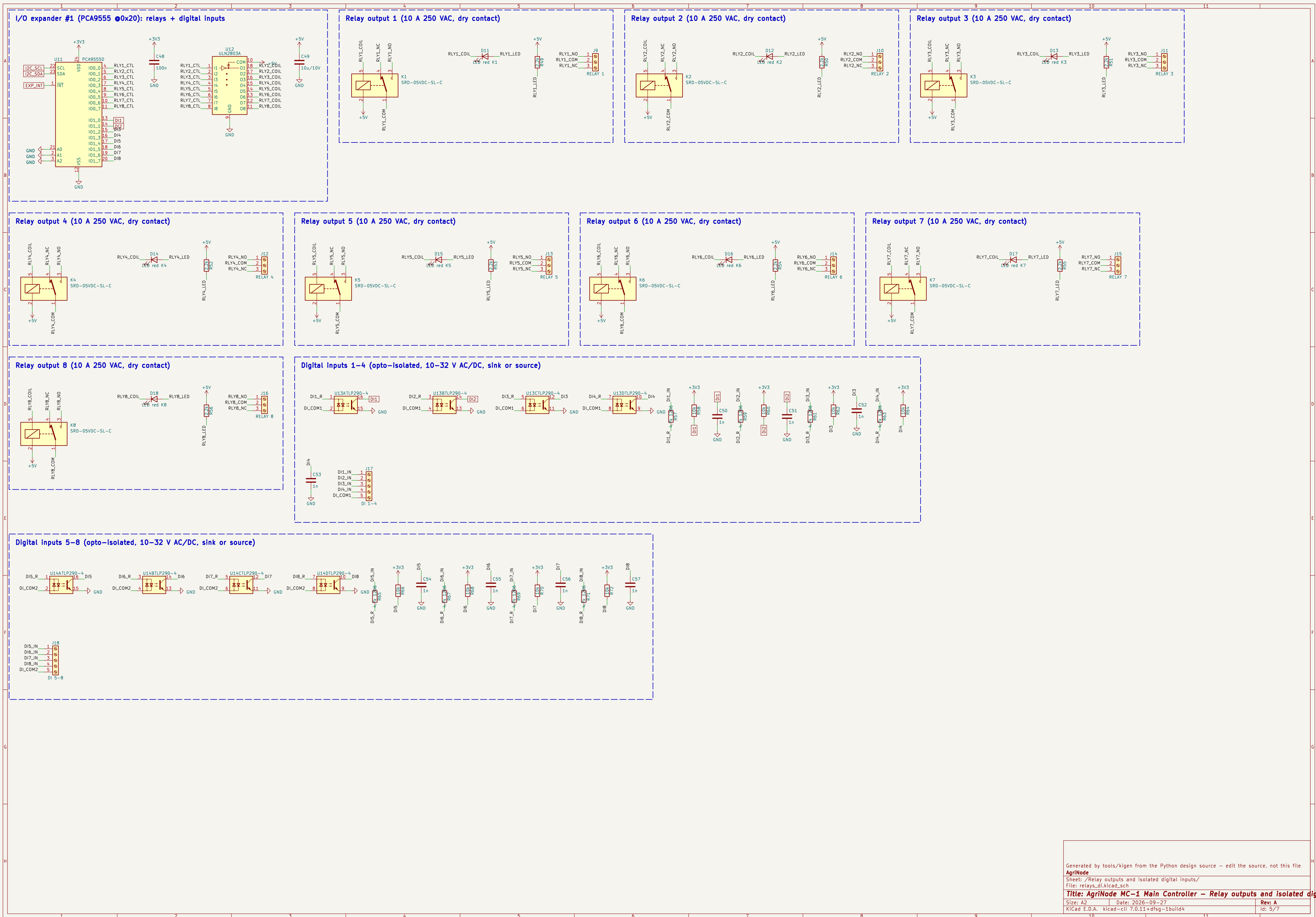
USB 5 V can power logic for programming (relays need main supply)



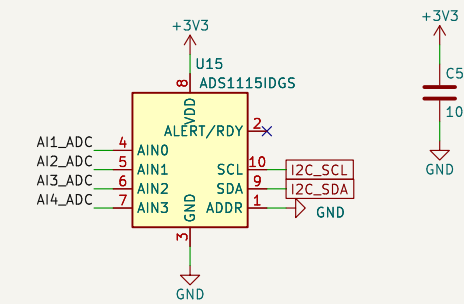
check against WIZnet W5500 reference design



3.3 V logic; card makes its own 3.8–4.2 V from +5V

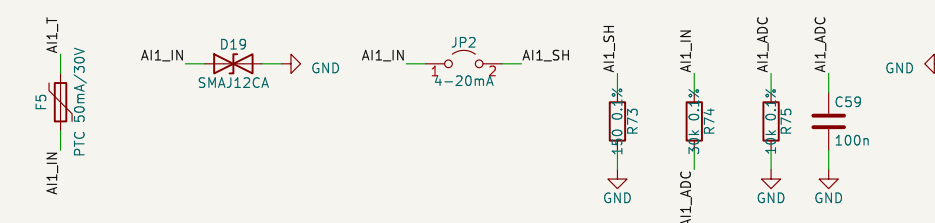


16-bit ADC (ADS1115 @0x48)



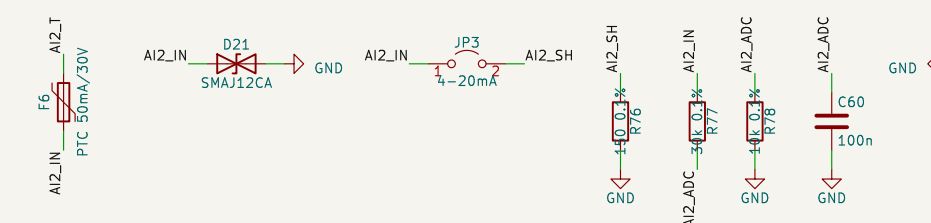
Analog input 1: 0–10 V or 4–20 mA (jumper)

V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



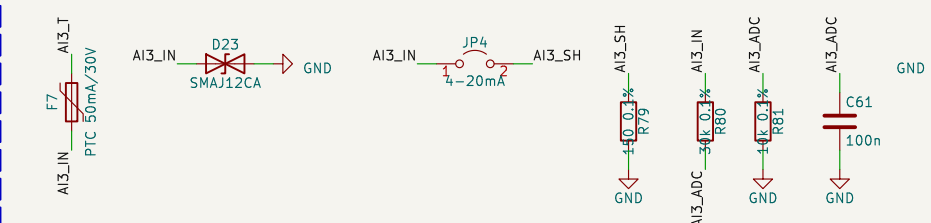
Analog input 2: 0–10 V or 4–20 mA (jumper)

V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



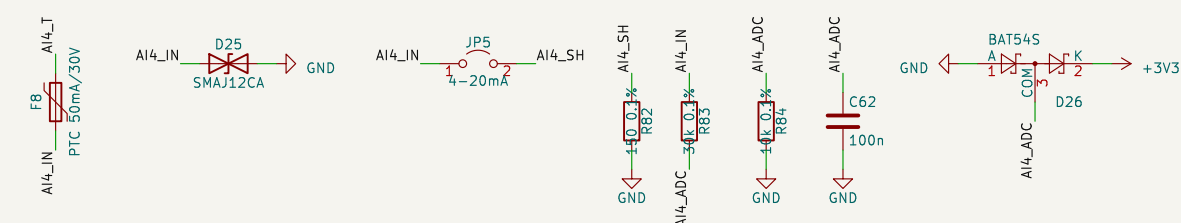
Analog input 3: 0–10 V or 4–20 mA (jumper)

V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V

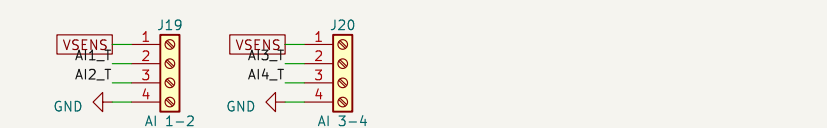


Analog input 4: 0–10 V or 4–20 mA (jumper)

V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V

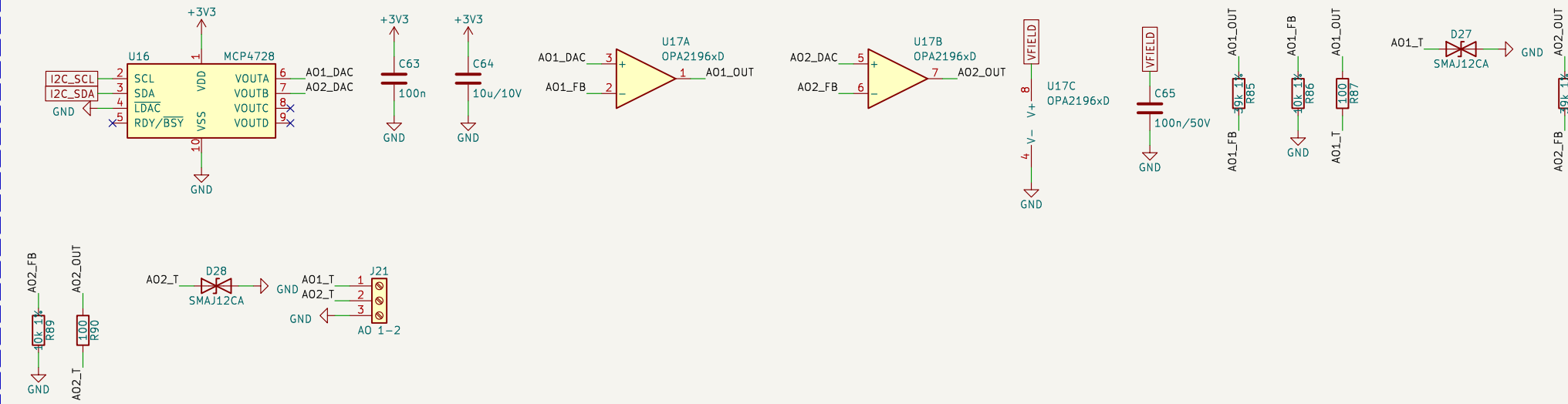


Analog input terminals (VSENS = loop power for 2-wire transmitters)

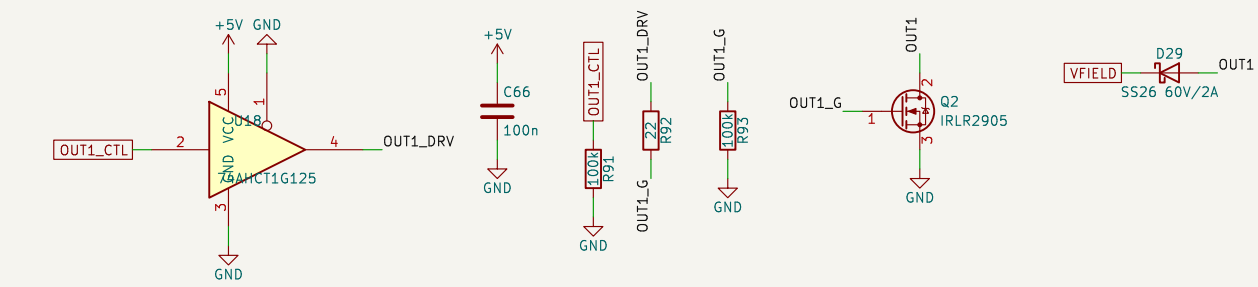


Analog outputs 0–10 V (MCP4728 + OPA2196, gain 4.9)

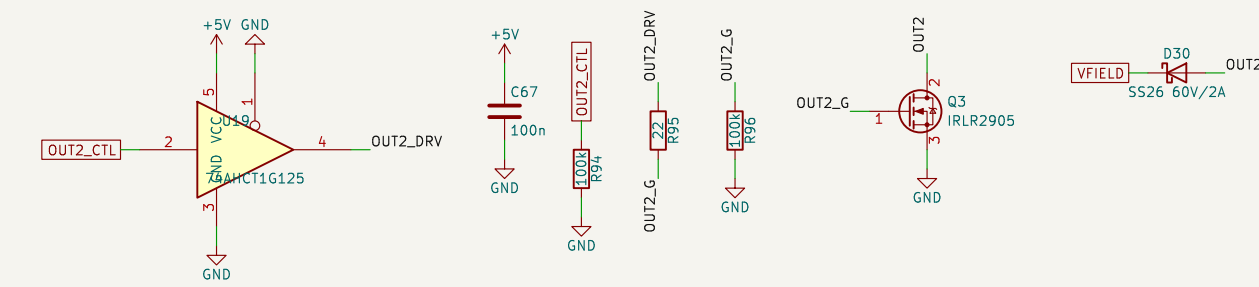
needs $V_{SYS} \geq 12\text{ V}$ for full 10 W



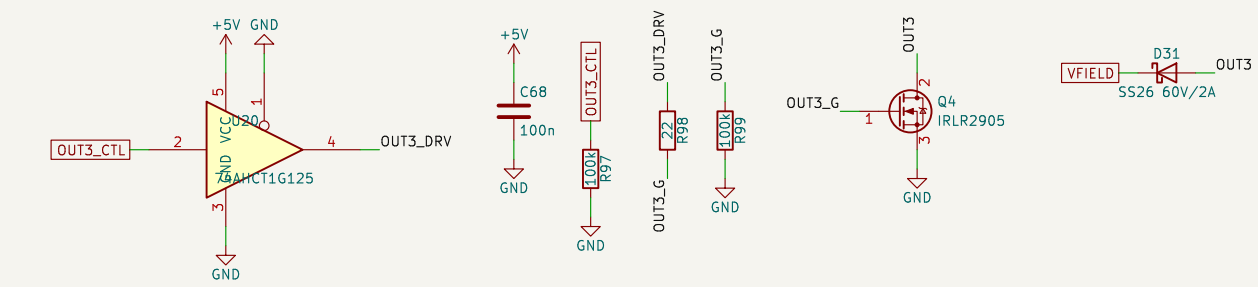
MOSFET output 1 (low side, PWM, up to 2 A)



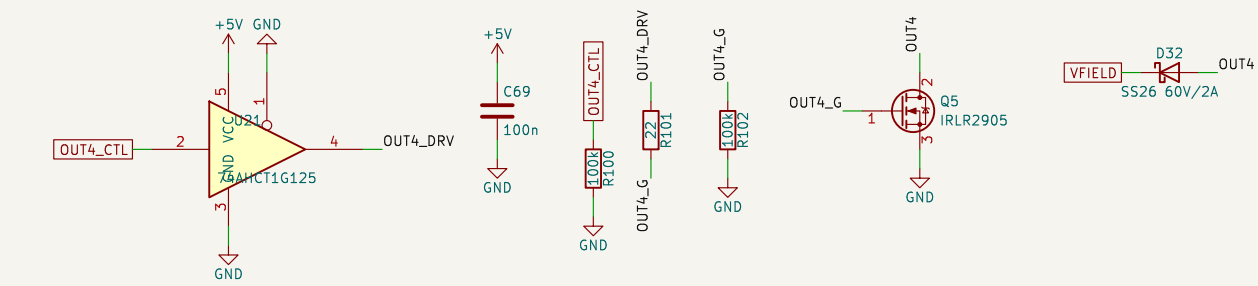
MOSFET output 2 (low side, PWM, up to 2 A)



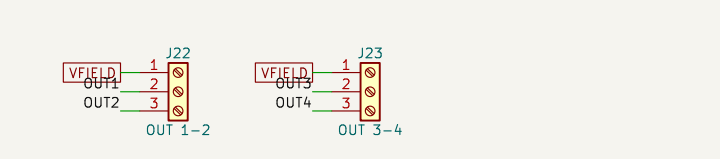
MOSFET output 3 (low side, PWM, up to 2 A)



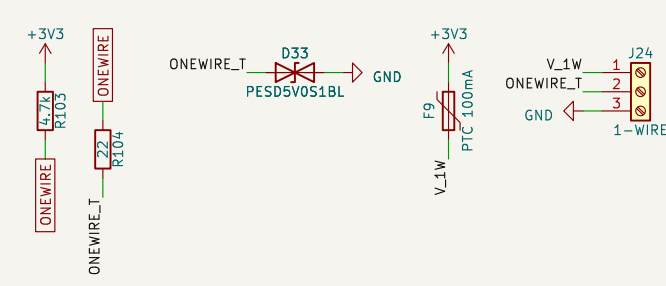
MOSFET output 4 (low side, PWM, up to 2 A)



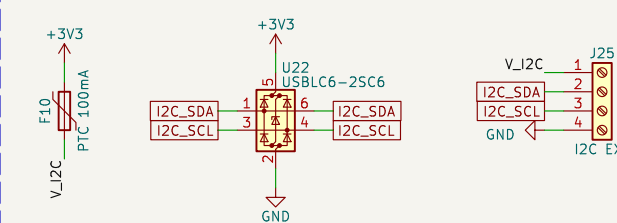
MOSFET output terminals (load between VFIELD and OUTx)



1-Wire bus (DS18B20 probes)



External I2C sensor port (SHT4x, SCD41, BH1750 ...; short cables)



[illegible]

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AgriNode		
Sheet: /RS-485 buses/ File: fieldbus.kicad_sch		
Title: AgriNode MC-1 Main Controller – RS-485 buses		
Size: A3	Date: 2026-09-27	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7